

Marn — Product Blueprint

Business thesis, four surfaces, measurement system, architecture, regulatory posture, roadmap

Marn — companion app for physical rehab and pain relief

MARN

Technical and Product Blueprint

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How to read this

This document has two audiences and does not apologise for either.

An investor should be able to read Parts One and Two and understand what the business is, why it is defensible, and what the money buys. An engineer — or a coding agent working from this file — should be able to read Parts Three through Six and build the thing without asking a question that isn't already flagged as open.

Where the two audiences want different things, the engineering detail wins and the prose carries the argument around it. Anything genuinely undecided is marked **OPEN** rather than papered over. A blueprint that pretends to certainty it doesn't have is worse than no blueprint, because it hides the places where the plan will bend.

Name. Marn — from مرن, Arabic for supple or flexible. Short, ownable, works in both scripts. Trademark clearance is outstanding and is the first item on the legal checklist.

PART ONE — THE BUSINESS

1. Executive summary

1.1 The one-sentence version

Marn is a companion app for physical rehab and pain relief: assisted stretching and physical recovery delivered one-to-one in studios, tracked in an app a member opens every day, so the relief they feel is backed by a number they can watch move over months.

1.2 Why the framing matters

There is an obvious version of this business and a better one.

The obvious version is a stretch studio members visit when something hurts, with a booking app bolted on. Hire flexologists, fit out a unit in a mall, sell packages. It works, it is being done, and it has no defensible position: anyone with capital can buy the same beds and hire from the same pool. The app is a booking form. Competition is on location and price, and price competition in a service business with fixed labour costs ends badly.

The better version inverts it. **Pain relief and physical rehab are what a member is buying; the app is where they live with it, and the studio is where the physical work happens.** A member opens the app between visits, not just to book — to see what hurt before and after, whether it's actually easing, and what to do at home. Every member has a longitudinal record behind that experience — ten joint angles in degrees, captured every six to eight sessions, tied to what was worked and how they slept — and that record is what turns "I feel a bit better" into something a member can see and trust. Nobody in this market gives a member that. It is the reason members stay past month three, the reason a corporate buyer signs, and the thing an acquirer is actually buying.

The practical consequence for engineering: **the app a member returns to is the core system, and the assessment pipeline behind it is what makes its pain-relief claims trustworthy rather than anecdotal. Booking and payments are commodity plumbing to be built as thinly as possible and never lovingly.**

1.3 What we are building

Three surfaces on one backend.

Surface	Who	Where
Member app	Paying members	iOS, Android, mobile web
Coach console	Flexologists	Tablet at the bed, laptop at the desk
Studio manager console	Studio managers	Laptop at the desk, tablet on the floor
Corporate portal	Employer accounts	Desktop web

Coach console and studio manager console started as one surface for two roles; the Phase 1 root build (starting 2026-08-11) splits them into two consoles with different capabilities, not just a shared screen with a role flag — see `docs/adr/0008-studio-manager-role.md`.

Behind them: a Postgres database whose central table is one row per muscle group per assessment, a scoring engine, an integration layer for the BodyMap measurement device and consumer wearables, and an operations layer covering scheduling, resources, credits and notifications.

1.4 Service model

One-to-one only. No group formats in the first year — they dilute the measurement story and change the staffing model.

Service	Length	Price
Assisted Stretch	30 min	AED 100
Assisted Stretch — Long	60 min	AED 190
Compression Recovery	30 min	AED 90
Oxygen Reset	20 min	AED 80
Hydration / nutrition add-on	—	AED 25–35

Prices are working figures. What matters architecturally is the *shape*: a base session, longer variants, equipment-led sessions that consume a resource rather than a coach, and small attachable add-ons. The booking engine is built

against that shape, so a pricing change is a row in a table.

1.5 Positioning — and the word we do not use

Marn is a **wellness and recovery studio**. It is not a clinic, and the distinction is load-bearing.

A clinic is regulated by the Dubai Health Authority, requires a licensed facility, requires licensed physiotherapists rather than trained flexologists, and carries a different insurance and record-keeping regime. That is a viable business but a slower and more expensive one, and it is not what we are building first.

Pain relief and physical rehab are the headline benefit we market — that's the whole point of the product and it is not the thing this section restricts. What's restricted is *how we describe delivering it*: no diagnosing the cause of pain, no treatment plans, no promise to cure. We describe measured change — "hip flexion up 9°, reported pain down from 6 to 3" — and let the member draw their own conclusion. Selling the outcome and staying out of clinical language are not in tension; the copy just has to do the first without doing the second.

The consequence runs all the way into the codebase and is not merely a marketing preference:

- Staff are **coaches** or **flexologists**. Never *therapists*.
- Customers are **members**. Never *patients*.
- Sessions produce **assessments** and **observations**. Never *diagnoses* or *treatment plans*.
- Scores are **fitness metrics**. The app says *below your target range*, never *your recovery is poor*.
- Readiness screening is a **referral gate**, not a liability form. Red flags route people to a doctor.

That last point deserves emphasis because it is the one that is both ethically right and legally protective. A member reporting acute pain, recent surgery, neurological symptoms or cardiac history should be blocked from booking and told to see a physician. Building that as a hard gate rather than a soft warning is the clearest possible evidence that we are not practising medicine — and it is simply the correct thing to do to someone who has walked in unwell.

1.6 Constraints shaping every decision

1. **One engineer.** A solo CTO working with a coding agent. Every architectural choice is measured against whether one person can operate it at two in the morning.
2. **Three months to studio opening.** The coach console must be usable on day one because it is the only piece the business cannot run without.
3. **Data residency.** Health data generated in the UAE should be hosted in the UAE. The prototype is not, and must never hold a real member's record.
4. **Future formats.** Container pods at golf courses and sports facilities are a stated phase-three ambition, which makes offline-capable capture a design requirement now rather than a retrofit.

2. The opportunity

2.1 Market

The UAE, and Dubai in particular, is unusually well suited to an app-led pain-relief and rehab business:

- **A young, high-income, health-engaged expatriate population.** The overwhelming majority of Dubai's residents are expatriates, disproportionately professional and disproportionately willing to spend on preventative health.
- **A large and visible endurance and fitness scene.** Runners, cyclists, Hyrox competitors, padel and golf. These are people who already own wearables and already think in numbers, which means the core product needs no education.
- **Desk-bound professional work in a car-dependent city.** Hip flexor shortening and thoracic stiffness are close to universal in the target segment.
- **Corporate wellness budgets that exist and are looking for something to buy.** Most current spend goes on gym subsidies with no measurable outcome. A per-employee mobility report is a materially easier sell than a discount code.

2.2 Segments, in commercial priority order

Professionals, 30–55. The core. They come for pain, stay for the graph. Highest volume, most predictable, most referable. Everything in the product is designed for this person first.

Performance athletes. Lower volume, higher frequency, highest word-of-mouth value. They care about range-of-motion numbers more than anyone and will use the wearable integrations properly. Disproportionate marketing value relative to revenue.

Longevity and 50+. Growing fast, high willingness to pay, longest retention. Requires the most careful handling: this is where readiness screening earns its keep and where clinical-sounding language would be most dangerous.

Corporate accounts. The multiplier. One signed employer replaces months of individual acquisition, and the aggregate anonymised reporting is only possible because of the measurement layer. Phase four, but designed for from the schema up.

2.3 Why tracked progress is the wedge

Three things follow from a member being able to see their own relief measured, that do not follow from a stretch session alone:

Retention. Service businesses churn because the customer cannot tell whether it worked. A member looking at a hamstring arc that has moved from 52° to 71° over four months does not churn the way someone with a half-used class pack churns.

Referral. A before-and-after progress card is shareable in a way that "I go to a stretch place" is not. This is the cheapest acquisition channel available and it is a direct product of the data.

Enterprise. "Your engineering team's average thoracic rotation improved 14% over the quarter, and reported back pain fell from 38% to 21%" is a renewal conversation. "Your staff used 340 sessions" is not.

2.4 Honest risks to the thesis

The measurement story only works if the numbers are trustworthy. Range-of-motion measurement has real inter-rater variability — two coaches measuring the same hamstring can differ by several degrees. If members see their score bounce because a different coach measured them, the entire proposition collapses.

This is the single biggest product risk in the business and it is a protocol problem before it is a software problem. Mitigations are covered in §5.5, but stated plainly here: **standardised measurement protocol, device-first**

capture wherever possible, and score presentation that shows trend more prominently than any single reading.

3. Product vision

3.1 The member's year

Week one. They come in with a specific complaint — lower back after long drives, shoulder that will not go overhead. They complete a readiness screening. A coach measures ten muscle groups. They leave with a first assessment, a plain-language summary of what was found, and three home stretches.

Month one. Four sessions. The second assessment shows small movement. The app has stopped being empty and has started being a record. They have done their home programme perhaps half the time, and the app knows.

Month three. The graph now has a shape. Some groups have moved a lot and some have not, and the app says so in both directions — *your hip flexors have gained 6°, your thoracic rotation hasn't moved and that's now the limiter*. They have hit a consistency milestone. They have added one friend from the gym.

Month six. They are a different customer. They book without prompting, they have referred someone, and the reason they do not stop is that stopping means watching the line go back down. Their employer has started asking what they are doing.

That arc is the product. Everything in the feature list either serves it or is deleted.

3.2 The coach's shift

A coach arrives and opens the console. Today's floor is a list with times, names and services. Two members have checked in from their phones: one slept badly and flags a stiff shoulder, one is fine. One member carries a safety flag about a shoulder impingement, shown before the coach touches them.

Between sessions, the coach records what was done — modalities, duration, perceived exertion, pain before and after — and writes two things: a clinical note for colleagues and a plain-language summary for the member. The summary is mandatory. It is not paperwork; it is the product the member paid for.

Every sixth to eighth session the coach re-measures, either by importing from the BodyMap device or by typing ten numbers. The member's app updates while they are still in the building.

The design constraint that matters: if the console is slower than a paper notebook, coaches will use the notebook and the dataset will never exist. Speed on the floor outranks every other consideration in that surface.

4. Feature specification

Each feature carries a phase marker: **P1** pilot, **P2** launch, **P3** depth, **P4** scale. The roadmap in §11 sequences them.

4.1 Member app

4.1.1 Scores and progress — strengths and weaknesses (P1/P2)

Four composite scores, each 0–100, each with its inputs visible on tap:

Score	Meaning
Flexibility	Static range across all measured groups as a percentage of target arc
Mobility	Weighted toward the joints that gate real movement — hips, thoracic, shoulders, ankles
Recovery	Readiness: adherence, wearable signal, recent exertion, attendance
Consistency	Attendance and home-programme completion over a rolling window

The progress view must present **both sides of the ledger**. Early versions of products like this show only deficits, which is demoralising and, worse, inaccurate — a member who has gained 9° in hip flexion while their calves stayed flat has a success and a problem, and the app should name both.

Concretely, the progress screen shows:

- **What's working** — the two or three groups with the largest gain since the previous assessment, named with the number.
- **What's limiting you** — the groups furthest from target arc, with an explanation of what that restricts in daily life, not just a percentage.
- **What hasn't moved** — groups that are neither improving nor severely restricted. This is the most actionable category and the one everyone forgets to surface.
- **Trend before value**. The chart is the primary object; the current number is secondary. This is partly a design preference and partly the mitigation for measurement variance described in §2.4.

4.1.2 Whole-body range-of-motion map (P1)

Front and back anatomical figure, ten muscle groups coloured against target arc on a four-band scale (restricted, limited, optimal, excellent). Tapping a group gives current degrees, target, percentage of arc, drift since last assessment, and a plain-language note about what that joint governs.

OPEN: the prototype uses a simplified geometric figure. A properly drawn anatomical SVG is a design task, not an engineering one, and should be commissioned before launch.

4.1.3 Booking with live equipment availability (P2)

Booking is modelled on **resources**, not on services. A resource is a coach, a stretch bed, a compression rig, an oxygen chamber, or the immersive sound room. A booking reserves one or more.

This matters because the studio sells things that are not coach-time. Compression boots and the sound room are capacity-limited assets with their own utilisation curve, and a member should be able to see *"boots free at 18:30"* the same way they see coach availability. Treating them as bolt-on extras rather than first-class bookable resources is the modelling error that would force a rewrite in month six, and it also under-sells genuinely differentiating equipment.

The same abstraction gives us, for free: multi-resource bookings (stretch then boots), equipment-only sessions with no coach, room-constrained scheduling, and eventually home-service dispatch, which is just a resource with a travel radius.

4.1.4 Community — friends and shared progress (P3)

Members can add friends and see each other's progress. This is a genuine retention mechanic in a category where accountability is most of the battle, and the gym-partner dynamic already exists offline.

It is also the feature most capable of causing real harm, and needs designing accordingly.

Range-of-motion data, pain scores and safety flags are health data. A member's friend list is not a lawful basis for disclosing it. The design rules are therefore non-negotiable:

- **Opt-in per member and per field.** Sharing is off by default. Turning on "share my progress" is a deliberate, explained action.
- **Scores and milestones only.** Friends see composite scores, streaks, milestones and session counts. Friends never see per-muscle clinical detail, coach notes, pain scores, readiness answers, or safety flags. These are separate database reads and separate permission checks, not a UI hide.
- **Mutual consent, and revocable.** A friend link requires acceptance by both parties and either can sever it, which immediately and retroactively removes access.
- **No leaderboards by default.** Ranking strangers by flexibility is a bad idea for a population that includes people in pain and people with body-image sensitivity. Friend comparison is opt-in; global ranking is not built.
- **No implicit disclosure through absence.** A member who stops attending should not have their disappearance broadcast.

4.1.5 Milestones (P3)

Consistency and target-based milestones: session counts, attendance streaks, home-programme streaks, first time a muscle group reaches its target arc, largest single-assessment gain, anniversary markers. Shareable as a progress card, which doubles as the referral engine.

Safety design, and this is not boilerplate. Streak mechanics applied to physical training can push people to train through pain or injury. Three rules:

- **Rest days do not break streaks.** Consistency is measured against a prescribed weekly cadence, not against consecutive days. A programme that says "four times a week" counts a streak on four.
- **Pain suppresses nudges.** A member whose last check-in reported high pain, or who carries an active safety flag, receives no streak-preservation prompts. The system should never nag someone who has just told it they are hurting.
- **Milestones celebrate range and consistency, never intensity.** No badge for a higher RPE. Nothing in the product should reward pushing harder.

4.1.6 Home programme and adherence (P2)

Coach-prescribed self-stretches with descriptions, target durations and video. Members mark completion; completion feeds the consistency and recovery scores. Idempotent per day.

4.1.7 Pre-session check-in (P2)

Two taps before arriving: pain areas tapped on a body diagram, sleep quality, current pain level, optional note. Surfaces on the coach's console before the member walks in. Extremely cheap to build, disproportionate effect on perceived personalisation.

4.1.8 Session history and summaries (P1)

Every completed session, with the coach's plain-language summary. This is the highest-value screen in the app relative to its build cost.

4.1.9 Wearable connection (P3)

Whoop, Apple Health, Garmin, Oura. Sharpens the recovery score from an estimate into a measurement. Explicit consent, scoped, revocable. See §9.2 for the residency complication.

4.1.10 Readiness screening (P1)

PAR-Q-derived questionnaire, completed with a coach at first visit. Cleared members can book; uncleared members cannot. Red-flag answers produce a referral message, not a workaround.

4.2 Coach console

As of the Phase 1 root build (2026-08-11), booking confirm/decline/reassign (§4.2.1, §4.2.2) moved to the studio manager console, not the coach console — a coach's schedule is read-only against already-approved bookings, and a coach's member-record access excludes contact and payment data. See `docs/adr/0008-studio-manager-role.md` for the full reasoning; not rewriting §4.2.1/§4.2.2 below wholesale since they still describe the feature correctly, just not on this console.

4.2.1 Floor schedule (P1)

Today's bookings by time with member, service, resources, coach and status. Confirm, decline, reassign, mark arrived. Tablet-first layout; must work on a laptop.

4.2.2 Request inbox (P1)

Pending booking requests with member context and any safety flags visible before acceptance. Assign a coach on confirm. Member is notified by push and WhatsApp.

4.2.3 Assessment capture (P1)

Three routes in — BodyMap device import, file import, manual entry — all landing on one schema. Manual entry is always available, works offline, and is the pilot's primary path. Previous values shown alongside for sanity-checking, with implausible jumps flagged for confirmation rather than silently accepted.

4.2.4 Session logging (P1)

Modalities used, duration, RPE, pain before and after, internal clinical note, and the member-facing summary. The summary field is **required** — the API rejects a session without one. Decrements credits, advances streak, closes the booking, recomputes scores.

4.2.5 Safety flags and contraindications (P1)

Persistent per-member flags surfaced on the schedule, in the request inbox, and at the top of the member record. Created by coaches, cleared explicitly, timestamped and attributed.

4.2.6 Programme prescription (P2)

Assign a home block from a template library or build one. Adherence flows back automatically.

4.2.7 Coach outcome metrics (P3)

Sessions delivered, average pain reduction, average range gained per member. This is the number that manages quality, informs progression and compensation, and — aggregated and anonymised — sells a corporate contract. It is also politically delicate inside a small team and should be introduced as a coaching tool before it is ever a performance-management one.

4.2.8 Capacity and utilisation (P3)

Seven-day utilisation by day and by resource. Utilisation is the entire economics of a studio business; anything under 40% is inventory to be sold at short notice, which leads directly to the waitlist feature.

4.2.9 Waitlist and short-notice fill (P3)

When a slot frees inside a threshold, notify nearby members with matching preferences at a discount. The single highest-ROI feature in the operational set, and only possible because we hold member location preferences and booking patterns.

4.3 Corporate portal (P4)

Company administrator account; employee invitation and verification; pooled or per-employee credit allocation; usage reporting; and **aggregate anonymised mobility reporting** with a minimum cohort size (no report on a group small enough to identify an individual — 15 is a reasonable floor). Employers see population trends. Employers never see an individual's data, and the product should say so loudly, because the fear that it might is the main objection to enterprise wellness tooling.

4.4 Administration (P2/P3)

Service and price management, resource and site management, coach roster and permissions, credit and package administration with a full audit trail, and content management for the home-programme library.

PART TWO — THE MEASUREMENT SYSTEM

5. Measurement, scoring and the device boundary

5.1 Why this chapter comes before architecture

Most of this system is unremarkable. Booking, payments, notifications and rostering are solved problems and the implementation choices barely matter. The measurement layer is the part that is specific to this business, the part

that is hardest to change later, and the part that determines whether there is a company here at all. It gets designed first.

5.2 The canonical model

Three entities, and the relationship between them is the whole design.

Assessment — one capture event. Who was measured, by whom, when, by what method, on what device.

Measurement — one number about one thing: a muscle group, a joint angle in degrees, and the target for that group. Ten per assessment today; the schema does not care if it becomes thirty.

Target — the reference arc a group is measured against. Currently a global constant per group. It should eventually be adjusted for age and sex, because a 58-year-old member measured against a 25-year-old's normative range is being told something both discouraging and untrue. **OPEN: normative reference data source and adjustment model.**

Measurements are stored as **individual rows, not as a JSON blob on the assessment**. This is the single most consequential schema decision in the document. A blob is faster to write today and makes the question that justifies the entire company — *show me left hamstring across every member over eighteen months, segmented by age* — an unindexable string-parsing exercise. One row per measurement, indexed on member and muscle key and date.

5.3 The ten groups

Key	Group	Region	Target
hamstrings	Hamstrings	Lower	90°
hip_flexors	Hip Flexors	Lower	20°
quadriceps	Quadriceps	Lower	140°
glutes	Glutes	Lower	30°
calves	Calves	Lower	20°
lower_back	Lower Back	Core	60°
thoracic	Thoracic Spine	Core	45°
shoulders	Shoulders	Upper	180°
chest	Chest (Pecs)	Upper	45°
neck	Neck	Upper	80°

OPEN: bilateral capture. Left and right differ, sometimes substantially, and asymmetry is clinically more interesting than absolute range. The schema carries a `side` field from day one but the pilot captures a single value per group to keep assessment under ten minutes. Revisit once the BodyMap output format is known — if the device reports bilaterally, capture bilaterally.

5.4 Scoring

All scoring lives in one file with no database and no UI imports, because it will be rewritten once a physiotherapist has opinions about the weights, and that rewrite should touch nothing else.

Flexibility — mean of `degrees ÷ target` across all measured groups, as a percentage. Simple, explainable, and defensible to a member who asks how it is calculated.

Mobility — weighted toward the joints that gate compound movement: thoracic 0.30, hip flexors 0.25, shoulders 0.25, calves 0.20. These weights are a judgement call and should be treated as provisional.

Recovery — a blend of home-programme adherence, wearable linkage, recent perceived exertion and attendance streak. The wearable term is currently a flat bonus for having a device connected, which is a placeholder for real HRV and sleep. When a real signal arrives, the function's shape does not change.

Consistency — attendance against prescribed cadence plus home-programme completion over a rolling 28 days. Deliberately separate from Recovery so that a member can be consistent without being well-recovered, and vice versa.

Presentation rule: every score shows its inputs on tap. A number a member cannot interrogate is a number they will not trust, and trust in these four numbers is the whole business.

5.5 Measurement reliability

§2.4 named this as the largest product risk. The mitigations:

1. **Written protocol per group** — member position, stabilisation, landmark, end-point definition. Laminated at each bed. This is a training artefact, not software, and it is the highest-leverage thing in this section.
2. **Device-first.** BodyMap capture wherever available; manual entry recorded as a distinct `source` so that variance can be analysed by method later.
3. **Same-coach re-test where scheduling allows,** recorded either way via `coachId` on the assessment.
4. **Implausible-delta confirmation.** A change beyond a configurable threshold prompts the coach to confirm rather than silently writing an outlier.
5. **Trend over value in the UI,** so a single noisy reading does not read as regression.
6. **Eventually: a variance dashboard** measuring inter-coach spread on the same member. Once there is enough data, this becomes a training tool.

5.6 The BodyMap boundary

BodyMap is a third-party measurement device that reports mobility and flexibility figures to coaches. **Its integration surface is currently unknown**, which makes it the largest technical unknown in the build.

The response is an anti-corruption layer. No BodyMap data format touches the database. Everything lands on `NormalisedMeasurement[]` through an adapter, and three adapters are planned:

Adapter	Status	Purpose
<code>fromDeviceApi</code>	Stub, signature fixed	If the vendor exposes REST or webhooks
<code>fromExportFile</code>	Stub, signature fixed	CSV/JSON dropped by a studio-side agent
<code>fromManualEntry</code>	Implemented	Coach types the numbers. Always available

The adapter drops unrecognised muscle keys rather than guessing at them. Silent coercion of unknown data is how a measurement dataset quietly becomes worthless.

Manual entry being permanently first-class is not a fallback grudgingly retained. It is what lets the pilot run before any integration exists, what keeps a container pod working with no connectivity, and what makes a second measurement vendor a one-file change.

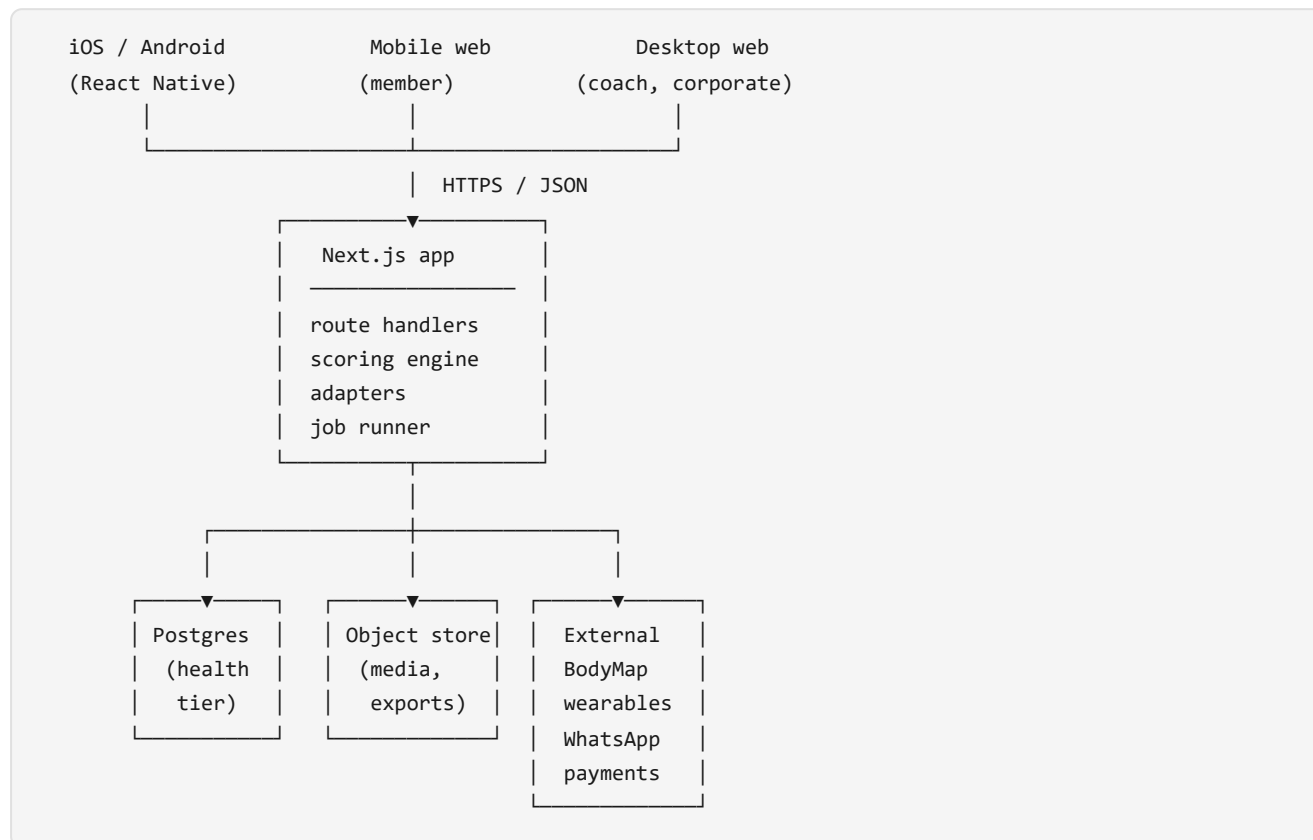
What we need from the vendor, in priority order: a sample export file; the technical contact; whether the device exposes an API or writes to a local database; whether it reports bilaterally; and what identifier it uses for a person, since that determines whether we can match records automatically or need a manual link step.

PART THREE — ENGINEERING

6. System architecture

6.1 Shape

One deployable. A Next.js application serving the member web surface, the coach console, the corporate portal and the API, backed by a single Postgres database. The mobile app is a React Native client against the same API.



This is deliberately unfashionable. There is no service mesh, no message broker, no separate API gateway, no microservices. One person cannot operate a distributed system and should not try. Every component removed

from this diagram is a component that cannot page you at 3am.

6.2 Why one deployable

Splitting the API from the web app would buy independent scaling we do not need and cost a network hop, a second deploy pipeline, a CORS configuration and a duplicated type definition. The mobile app is a separate client, not a separate service, and it consumes the same routes the web app does.

6.3 Background work

Jobs — notification dispatch, nightly score rollups, re-test reminders, waitlist fills, wearable polling — run through **pg-boss**, a queue that lives inside Postgres. This deletes Redis from the operational surface entirely. It is not the fastest queue available; it is the one that adds zero infrastructure and can be inspected with SQL.

6.4 Data tiering

Two logical tiers with different residency requirements:

Health tier — members, assessments, measurements, sessions, flags, check-ins, programmes, scores. UAE-resident. Never leaves.

Operational tier — product analytics, marketing attribution, error telemetry, CRM. May live anywhere, and must be scrubbed of health data before it does. Practically, this means error reporting is configured to strip request bodies by default rather than allow-list them, because the failure mode of the opposite choice is a member's pain score sitting in a foreign log aggregator.

6.5 Offline capability (P4, designed for now)

Container pods at golf courses and sports facilities will lose connectivity. The coach console must capture an assessment and a session with no network and sync later.

The design: assessment and session writes are **client-generated UUIDs queued locally and replayed**. They are append-only and effectively conflict-free — two coaches do not edit the same session. Booking is *not* offline-capable, because double-booking a resource is a real conflict with no clean automatic resolution; a pod with no connectivity accepts walk-ins and reconciles later.

Building this in phase four is acceptable. Designing the write path as append-only and client-ID-generated **now** is what makes phase four a feature rather than a rewrite.

7. Technology stack and decision log

7.1 The stack

Layer	Choice
Language	TypeScript, end to end
Repo	Turborepo monorepo

Layer	Choice
Web + API	Next.js (App Router), route handlers
Mobile	React Native via Expo
Database	PostgreSQL
Query layer	Drizzle ORM, migrations in repo
Jobs	pg-boss
Auth	better-auth, self-hosted, phone OTP primary
Object storage	S3-compatible
Styling	CSS custom properties, shared token file
Hosting (prototype)	Vercel + Neon
Hosting (production)	Containers, UAE region

7.2 Decisions and the alternatives rejected

TypeScript everywhere. One language across mobile, web, API and scripts. For a solo engineer this is not a preference, it is a survival requirement. Shared domain types mean a change to the measurement schema breaks compilation everywhere it is wrong, immediately.

Monorepo. *Rejected: separate repos per surface.* Separate repos mean a published shared package, a versioning dance, and drift. One repo, one commit, one truth.

Next.js as both frontend and API. *Rejected: NestJS or Fastify as a separate API.* A separate API is the right answer for a team of five. For one person it is a second deploy target and a second set of logs to correlate.

Expo for mobile. *Rejected: native Swift and Kotlin.* Two native codebases is two to three times the mobile work for UI that is overwhelmingly data display. The one place native depth is genuinely required is Apple HealthKit, which Expo handles with a config plugin and a development client.

Drizzle. *Rejected: Prisma.* Prisma is friendlier; Drizzle produces SQL you can read, which matters disproportionately when the interesting queries are analytical aggregations over the measurements table.

pg-boss. *Rejected: BullMQ with Redis, or a hosted queue.* Deletes an entire piece of infrastructure. Revisit only if job volume genuinely outgrows Postgres, which is many thousands of members away.

Self-hosted auth. *Rejected: Clerk, Auth0, Supabase Auth.* All three are excellent and all three put identity data on foreign infrastructure, which is an argument we would rather not have with a regulator about a health service. Phone OTP with a UAE SMS provider keeps identity in our database.

Postgres over a connection string, with no vendor extensions. *Rejected: Supabase's client libraries, PostgREST-from-the-browser, row-level-security-as-authorisation.* Every one of those is a good tool and every one welds the application to a vendor that has no UAE region. Authorisation lives in application code where it can be read, tested and audited.

Vercel and Neon for the prototype only. Chosen for speed to a shareable link. Neon specifically because its free tier does not pause a project into unavailability after a week of inactivity, which matters when an investor might open the link on a quiet Tuesday. Nothing Neon-specific exists outside one file.

7.3 The portability rule

Written down because it is the rule most likely to be violated under deadline pressure:

No vendor-proprietary primitive enters the codebase. The database is reached by connection string. The application builds to a standard container. Hosting and database are configuration, not architecture.

The cost of following this rule is a few days of foregone convenience. The cost of breaking it is discovering in month seven that moving to UAE hosting means a rewrite, at exactly the moment the first corporate client asks where the data lives.

8. Regulatory position, privacy and data residency

This chapter is written from a technical reading of public sources. It is not legal advice, and every material claim in it should be confirmed by UAE-qualified counsel before the first real member record exists.

8.1 The residency rule

UAE Federal Law No. 2 of 2019 on the Use of Information and Communication Technology in Health Fields restricts the storage, processing and transfer outside the UAE of health data generated in connection with health services provided in the UAE. Ministerial Resolution 51 of 2021 introduced specific exceptions, but the default position remains restrictive.

Whether a non-DHA-licensed wellness studio falls within scope is genuinely arguable — the law's operative definitions attach to health facilities, and we are deliberately not one. But the statute's scope language reaches services indirectly related to healthcare that handle electronic health data, and joint angles, pain scores and HRV are health data on any reasonable reading. Separately, the UAE Personal Data Protection Law treats health data as a special category regardless of who holds it.

The engineering position is therefore to comply as though the rule applies. The cost difference between hosting in the UAE and hosting elsewhere is negligible. The cost of migrating a live clinical dataset after a regulator or a corporate client's procurement team asks the question is not.

8.2 Consequences

- Managed platforms without a UAE region — Supabase, Firebase, Neon, PlanetScale, Vercel's own data products — are unavailable for the health tier in production. **They are fine for the prototype precisely because it holds no real data.** As of 2026-08-11, the root product also runs against Neon temporarily, for the same reason and under the same bound — no real member data yet. See `docs/adr/0010-neon-interim-production-database.md` for the trigger to migrate off it.
- Wearable aggregators such as Terra, Rook and Spike route through US or EU infrastructure. Using one is likely an export. Either integrate directly into the UAE backend or obtain counsel's sign-off.
- LLM features that touch health data need a UAE-resident inference endpoint. This constrains the auto-summary feature described in §9.5 and is the reason it is not in the pilot.
- Analytics and error reporting must be configured to exclude health data by default.

OPEN: final region. AWS `me-central-1` (UAE), Azure UAE North, and local providers are all candidates, and the choice is currently deferred. This is precisely why §7.3 exists — the decision can wait because it is configuration.

8.3 Consent

Separate, granular, revocable, logged:

Consent	Scope
Service	Assessment, session records, coach notes
Wearable	Named provider, named data types
Community	Sharing scores and milestones with accepted friends
Corporate	Inclusion in an employer's anonymised aggregate reporting
Media	Photographs and video, for progress records or marketing, separately
Marketing	Communications, entirely separate from all of the above

Every consent is timestamped, versioned against the policy text in force at the time, and revocable without losing access to the service. A member who withdraws community consent has their shared data withdrawn from friends' views immediately and retroactively.

8.4 Retention and subject rights

Wellness records, not medical records, so mandatory clinical retention periods do not apply — but members will ask for their data and occasionally for its deletion. Build export and deletion as real capabilities from the schema up: every table carries `memberId`, and a deletion routine that walks them is a day's work now and an archaeological dig later.

Deletion is anonymisation, not destruction, for aggregate purposes. A deleted member's measurements may remain in de-identified aggregate statistics; their identity, contact details, notes and history are removed. This must be stated plainly in the privacy policy rather than discovered.

8.5 Marketing claims

The wellness positioning constrains what may be said, in the product and outside it. No treatment claims, no cure claims, no pain-relief guarantees, no before-and-after imagery implying medical outcome. "Members improved average hip flexion by 14% over twelve weeks" is a factual statement about our own data. "Fixes back pain" is a medical claim and would invite exactly the regulatory attention the positioning exists to avoid.

9. Integrations

9.1 BodyMap

Covered in §5.6. Highest-priority unknown; three adapters, one canonical output.

9.2 Wearables (P3)

Whoop, Apple Health, Garmin, Oura. Apple Health is the awkward one: it is read on-device only, which means the mobile app extracts and forwards rather than the server polling an API. That is a feature, not a limitation — the data path runs from device to UAE backend with no third party in between.

Aggregators such as Terra, Rook and Spike would save perhaps two weeks of integration work per provider and would introduce a foreign processor into the health-data path. Start with direct Whoop and Apple integrations; revisit only with counsel's sign-off.

Scope requests to what the recovery score actually needs: heart-rate variability, sleep duration and quality, resting heart rate, daily strain. Not location, not workout GPS, not anything we cannot justify holding.

9.3 Notifications

Push via Expo to APNs and FCM. **WhatsApp Business API** through a business solution provider — Twilio, 360dialog, or a regional provider such as Unifonic. This is not optional in this market: a booking confirmation that does not arrive on WhatsApp has, functionally, not arrived. Email is for receipts and corporate reporting. SMS is the OTP fallback.

WhatsApp template approval takes days to weeks and is a common launch-blocker. Start that process in the first week of the build, not the last.

9.4 Payments (P2)

OPEN — deferred by decision. Candidates: Stripe, Network International, Telr; plus Tabby or Tamara for buy-now-pay-later on multi-session packages, which materially lifts average package size in this market. Apple Pay and Google Pay expected.

Whatever is chosen, the internal model is a **credit ledger, not a counter**. Packages, memberships, expiry, freezes, corporate-funded credits, refunds and gifted sessions are all entries in an append-only ledger with a derived balance. The alternative — a `sessions_remaining` integer that gets edited — is where booking systems' data integrity reliably dies, and the first billing dispute in month eight is when you find out.

9.5 Auto-generated session summaries (P4)

The coach records structured findings; a language model drafts the member-facing summary for the coach to approve. A real consistency win, deferred for two reasons: it sends health data to an inference endpoint and therefore needs a UAE-resident option (§8.2), and a coach's own words in month one are better training material than a model's.

9.6 Calendar and corporate identity (P4)

Calendar invitations for confirmed bookings. Single sign-on for corporate portal administrators. Both straightforward, both worth nothing until corporate accounts exist.

10. Security and access

10.1 Identity

Phone number as primary identifier, OTP authentication, optional Apple and Google sign-in. Separate identity domains for members and staff — a coach account is not a member account with a flag set, because that conflation is how privilege-escalation bugs are born.

10.2 Roles

Role	Access
Member	Own record only. Friends' shared scores where mutual consent exists
Coach	Assigned members at their site — check-in and session context needed for the session; assessment and session write; flag management. Not contact or payment details, and not booking approval as of the Phase 1 root build
Studio manager	All members at their site (coach access plus contact/payment); roster, shifts, capacity, credits and pricing; sole booking approval as of the Phase 1 root build
Corporate admin	Aggregate reporting for their organisation. Never individual records
Platform admin	Everything, with every action logged

The bolded exceptions were added 2026-08-11 — see [docs/adr/0008-studio-manager-role.md](#). Booking approval is no longer "coach, plus manager oversight"; it moved to the manager exclusively.

Authorisation is enforced server-side on every route against the authenticated identity. Client-side role checks are presentation only and are assumed hostile.

10.3 Audit

Append-only audit log covering: assessment created or amended; session logged or edited; flag raised or cleared; readiness status changed; credit adjusted; consent granted or withdrawn; member record exported or deleted; and any staff read of a member record outside their own site.

This is not a compliance checkbox. The first time a member disputes what a coach recorded, this log is the only thing standing between a conversation and a problem.

10.4 Practical controls

TLS everywhere. Encryption at rest. No health data in logs or error reports. Secrets in the platform secret store, never in the repository. Dependency scanning in CI. Rate limiting on OTP and booking endpoints. Backups encrypted, retained in-region, and **restore-tested quarterly** — an untested backup is a belief, not a backup.

11. Build roadmap

Sequenced as vertical slices. Each phase ends with something usable on the studio floor, because a solo engineer cannot afford a phase that produces only foundations.

Phase 1 — Pilot (weeks 0–8)

Goal: the studio can open and operate.

Schema and migrations · coach authentication · member roster and creation · readiness screening · assessment capture, manual first, with the BodyMap adapter stubbed · session logging with mandatory member summary · safety flags · scoring engine · member web portal, read-only — scores, body map, progress, session summaries · manual booking entry by staff.

Exit criterion: a coach can run a full day, and every member can see their results, without a spreadsheet existing anywhere in the building.

Phase 2 — Launch (weeks 8–14)

Goal: members self-serve.

Member authentication · self-service booking on the resource model · payments and the credit ledger · push and WhatsApp notifications · home programmes and adherence · pre-session check-in · mobile app, iOS first with Android within three weeks · cancellation policy.

Exit criterion: a member joins, books, pays, attends and reads their results with no staff intervention at any step.

Phase 3 — Depth (months 4–7)

Goal: retention mechanics and a second site.

Wearable integrations · community with the consent model in §4.1.4 · milestones with the safety rules in §4.1.5 · waitlist and short-notice capacity fill · capacity and coach-outcome dashboards · multi-site · re-test cadence reminders · referral tracking.

Phase 4 — Scale (months 7–12)

Goal: enterprise and new formats.

Corporate portal · offline capture and sync for pod sites · **migration to UAE-hosted production** · auto-generated summaries once a compliant endpoint exists · outcome analytics · Arabic and RTL.

Sequencing rules

- Coach console before member app, always. The console is operationally load-bearing; the app is not.
- No feature ships to members before it works on the floor.
- The UAE migration happens before the first corporate contract, not after it is signed.
- Arabic is deferred but never architecturally blocked: user-facing strings go through a lookup from day one, and layouts avoid direction-dependent assumptions.

12. Non-functional requirements

Performance. Member app cold start under three seconds on mid-range Android. Coach console interaction under 200ms for anything used on the floor. Assessment save under one second — a coach standing next to a member will not wait, and if they wait twice they will reach for paper.

Availability. 99.5% during studio hours, 06:00–22:00 GST. Degraded service outside those hours is acceptable. The console must fail soft: if the network drops, capture continues locally.

Scale targets. Year one: one site, roughly 500 active members, 40 sessions a day. Year three: four sites, roughly 4,000 members, 200 sessions a day, plus corporate accounts. These are small numbers and the architecture should not pretend otherwise — the measurements table reaches perhaps two million rows in three years, which is unremarkable for Postgres on modest hardware.

Environments. Local, preview per pull request, production. Preview branches get a throwaway database branch. Production migrations are reviewed and never automatic.

CI/CD. Type check, lint, unit tests on the scoring engine and adapters, build. Deploy on merge to main. Migrations run as an explicit, separate step.

Observability. Structured logs with correlation IDs and no health data. Error tracking with request bodies stripped by default rather than allow-listed. Uptime checks on the API and database. One dashboard — sessions logged today, assessments captured, booking failures, notification delivery, job queue depth — small enough to glance at daily and therefore actually looked at.

Testing. Scoring and adapters get real unit tests; they are pure functions with real consequences. The booking engine gets integration tests around resource conflicts and the credit ledger. The UI gets smoke tests on critical paths and nothing more — a solo engineer maintaining a large UI test suite is maintaining a test suite instead of a product.

13. Cost model

Infrastructure only. Excludes fit-out, equipment, salaries, rent and marketing.

Prototype and pre-launch: effectively zero. Free tiers throughout, plus one deployment platform subscription if the demo link needs commercial terms.

Year one, single site — monthly, USD:

Item	Estimate
Compute, UAE region	80–150
Postgres, managed, with backups	60–120
Object storage and CDN	15–30
WhatsApp Business API and templates	50–150
SMS OTP	30–80
Push notifications	0
Error tracking and uptime monitoring	30–60

Item	Estimate
Apple and Google developer accounts	~12 amortised
Total	≈ 300–600

Year three, four sites plus corporate: roughly 1,200–2,500 a month, dominated by messaging volume rather than compute. Messaging cost scales with member count; compute barely moves. If a line item surprises you later, it will be WhatsApp.

Non-infrastructure engineering costs to budget separately: anatomical illustration for the body map, the home-programme video library, penetration testing before the first corporate contract, and UAE legal review of the residency and consent position.

PART FOUR — RISK AND REFERENCE

14. Risk register

Ordered by expected damage, not by probability. Each carries an owner-level response rather than a generic "monitor".

14.1 Measurement variance destroys the proposition

Severity: critical. If members see their scores move because a different coach measured them, the entire premise fails and the failure is invisible until it is advanced. **Response:** written protocol per group (\$5.5), device-first capture, trend-dominant presentation, implausible-delta confirmation, and an inter-coach variance dashboard once data volume allows. Treat this as a training investment, not a software one.

14.2 BodyMap does not integrate

Severity: high. Probability: moderate. The vendor may expose no API, no export, or a closed desktop application. **Response:** already mitigated architecturally. Manual entry is first-class and is the pilot's primary path. Worst case is ninety seconds of coach time per assessment, which is a cost, not a blocker. Obtain a sample export and a technical contact early so the cost is known rather than assumed.

14.3 Regulatory reclassification

Severity: high. A regulator takes the view that assisted stretching with measurement constitutes a health service requiring DHA licensing. **Response:** the language discipline in §1.5 and the referral gate in §4.1.10 are the primary defence. Obtain a written legal opinion before opening, and structure so that becoming licensed is an upgrade path rather than a rebuild — the data model already supports licensed practitioners as a coach type.

14.4 Data residency enforcement

Severity: high. Probability: low near-term, rising with scale. Response: the portability rule (§7.3). Migration is a connection string and a container image. Complete it before the first corporate contract, since procurement will ask.

14.5 Solo engineering capacity

Severity: high. Probability: high. One person building three surfaces while a studio opens is the most likely source of schedule failure, and no architecture fixes it. **Response:** ruthless phase discipline. The console is the only thing that must exist at opening. Everything in Phase 3 and 4 is genuinely optional for a year. Identify, now, the point at which a second engineer or a contractor is hired, and what triggers it.

14.6 Coaches bypass the console

Severity: high. If the console is slower than paper, the dataset never accumulates and every downstream claim in this document evaporates. **Response:** floor-speed requirements in §12 are hard requirements. Test with real coaches in week three, not week twelve. Watch a coach use it in person before believing any of it works.

14.7 Community feature causes a privacy incident

Severity: high. Health data exposed to a friend, an ex-partner, or a colleague. **Response:** the consent model in §4.1.4. Separate reads, separate permission checks, opt-in per field, mutual and revocable links, no leaderboards. If this cannot be built properly, it should not be built.

14.8 Streak mechanics encourage training through injury

Severity: moderate, but reputationally severe if it happens. Response: rest days do not break streaks; pain suppresses nudges; nothing rewards intensity (§4.1.5).

14.9 Trademark collision

Severity: moderate. Probability: unknown until cleared. Response: clear "Marn" in UAE classes covering fitness and wellness services and software before any spend on signage, app store listings or domains.

14.10 Utilisation below break-even

Severity: moderate, and a business risk rather than a technical one. Response: the waitlist and short-notice fill feature exists precisely for this, and is the reason it is not treated as a nice-to-have.

14.11 Single-engineer key-person risk

Severity: moderate. One person holds the entire system in their head. **Response:** this document. Plus: no undocumented deploy steps, secrets held in a shared vault with recovery access, and a README that a competent stranger could follow.

15. Appendices

Appendix A — Data dictionary

coaches — id, name, initials, title, siteld, isDemo, createdAt

members — id, name, phone, goal, persona, joinedAt, credits, streak, wearable, parqCleared, parqAt, isDemo, createdAt

flags — id, memberId, text, since (*contraindications and safety notes*)

assessments — id, memberId, coachId, capturedAt, source `bodymap|manual`, deviceId, createdAt

measurements — id, assessmentId, memberId, muscleKey, degrees, target (*one row per group; the central table of the business*)

bookings — id, memberId, coachId, serviceId, date, time, status `requested|confirmed|completed|cancelled`, addons, aed, createdAt

sessions — id, memberId, coachId, bookingId, completedAt, mins, modalities, rpe, painBefore, painAfter, coachNotes (*internal*), memberSummary (*member-facing, required*), createdAt

programs — id, memberId, coachId, title, assignedAt, moves, completions

checkins — id, memberId, at, sleep, pain, areas, note

scoreDays — id, memberId, date, flexibility, mobility, recovery (*denormalised for chart reads*)

Not yet built: `sites`, `resources`, `resourceBookings`, `creditLedger`, `consents`, `auditLog`, `friendships`, `milestones`, `organisations`, `orgMembers`. Each is named here so the eventual migration is expected rather than discovered.

Appendix B — API surface

See `API.md` in the repository for the live contract. Current shape:

- **Read** — `GET /snapshot`, `GET /availability`, `GET /members/:id`, `GET /admin/tables`
- **Identity** — `POST /members`, `POST /coaches`, `POST /members/:id/parq`, `POST|DELETE /members/:id/flags`, `POST /members/:id/wearable`
- **Booking** — `POST /bookings`, `POST /bookings/:id/confirm`, `POST /bookings/:id/decline`, `DELETE /bookings/:id`
- **Measurement** — `POST /integrations/bodymap/import`, `POST /members/:id/assessments`
- **Session** — `POST /sessions` (*422 without a member summary*)
- **Programme** — `POST /members/:id/programs`, `POST /programs/:id/complete`
- **Intake** — `POST /checkins`

`GET /snapshot` returns the whole dataset and is a prototype convenience. It must be replaced with scoped, paginated endpoints before there is real volume; it is noted here so it is not inherited by accident.

Appendix C — Glossary

Adherence — whether a member completed the home programme they were prescribed.

Anti-corruption layer — a translation boundary. External data formats are converted to our schema at the edge, so a vendor change touches one file.

Assessment — one capture event containing many measurements.

Composite score — a single 0–100 figure derived from many measurements, so members have something simple to track.

Contraindication — a condition making a specific intervention unsafe for a specific person.

Credit ledger — append-only record of session credits bought, used, expired, frozen or gifted, with a derived balance.

Flexologist — a trained assisted-stretching coach. Not a licensed medical title, and the reason we never say *therapist*.

Goniometer — the instrument used to measure joint angle. The arc that appears throughout the product is drawn from its face.

HRV — heart-rate variability, the millisecond variation between heartbeats; the basis of most wearable recovery scores.

Longitudinal — data about the same person over time, as opposed to a single snapshot. The entire value of the product.

Modality — a technique or piece of equipment used in a session: assisted stretch, PNF, compression boots, oxygen, immersive sound.

PAR-Q — Physical Activity Readiness Questionnaire; the standard pre-exercise screening instrument.

PNF — proprioceptive neuromuscular facilitation, a contract-relax stretching technique.

Resource — anything a booking consumes: a coach, a bed, a compression rig, a chamber, a room.

ROM — range of motion, measured in degrees.

RPE — rate of perceived exertion, a 1–10 self-report of session intensity.

Target arc — the reference range of motion a group is measured against.

Appendix D — Build prompts for a coding agent

Each prompt assumes the repository, this document and `API.md` are in context. Run them in order; each should end with a working, committable state.

D1 — Schema and migrations

Extend `db/schema.ts` with the tables listed as *not yet built* in Appendix A: sites, resources, resourceBookings, creditLedger, consents, auditLog, friendships, milestones, organisations, orgMembers. Follow the existing conventions — text primary keys, explicit foreign key columns, jsonb for genuinely unstructured fields only. Generate a migration. Do not modify existing tables except to add a `siteId` to bookings and sessions.

D2 — Resource-based booking

Replace service-based availability with resource-based availability per §4.1.3. A booking reserves one or more resources. Add conflict detection per resource. Compression boots, the oxygen chamber and the immersive sound room become resources with their own capacity. Availability must return which resources are free, not just which times.

D3 — Credit ledger

Replace the `credits` integer on members with an append-only ledger per §9.4. Entry types: purchase, consumption, expiry, freeze, unfreeze, refund, gift, corporate grant. Balance is derived, never stored. Add integration tests covering double-spend, expiry at a boundary, and refund after consumption.

D4 — Authentication and authorisation

Add better-auth with phone OTP. Separate member and staff identity domains per §10.1. Enforce the role matrix from §10.2 server-side on every route. Add an authorisation test per route asserting that a member cannot read another member's record and a corporate admin cannot read any individual record.

D5 — Scoring v2

Add the Consistency score per §5.4. Make target arcs adjustable by age band, defaulting to current global values. Keep everything in `lib/scoring.ts` with no database or React imports. Unit-test each function against a fixture set including empty-measurement and single-assessment cases.

D6 — Mobile app

Add `apps/mobile` as an Expo React Native app consuming the existing API. Port the member surfaces from `components/Member.tsx`, sharing types from `packages/core`. iOS first. Apple HealthKit via a config plugin and development client. Reuse the design tokens from `app/globals.css`.

D7 — Community and milestones

Implement friendships and milestones per §4.1.4 and §4.1.5. Sharing is opt-in per field and off by default. Friends' scores and milestones are served from a dedicated endpoint that reads only shareable fields — do not filter a full member payload in the client. Rest days must not break streaks, and members with an active safety flag or a recent high-pain check-in receive no streak prompts.

D8 — Offline capture

Make assessment and session writes offline-capable per §6.5. Client-generated UUIDs, local queue, replay on reconnect, idempotent server handlers. Booking stays online-only. Add a visible sync state in the coach

console — a coach must always know whether their work has been saved.

Closing note

The largest risks in this plan are not technical. They are that measurement turns out to be noisier than hoped, that one engineer is one engineer, and that a studio has to open on a date regardless of what the software is doing.

The architecture is therefore chosen to be boring and portable, so that engineering effort goes into the two things that actually differentiate the business: a companion app whose pain-relief and progress claims are backed by a measurement dataset that can be trusted, and a coach console fast enough that coaches use it willingly.

Everything else in this document is negotiable. Those two are not.

*Marn — Technical and Product Blueprint · Version 0.1 · August 2026 Open items are marked **OPEN** throughout. Sections 8 and 14.3 require review by UAE-qualified counsel before the first real member record exists.*